

The basic command line syntax to invoke Nmap is as follows:

```
nmap [scan type(s)] [options] {target specification}
```

Nmap has a huge list of command line options, generally categorised into target specification, host listing, port specifications, service identification, scan technique, scripted scans and output options.

Some of the Nmap switches only work when run as the root (superuser). Let's look at some of the basic Nmap commands:

Command	Purpose
<code>nmap -sL 192.168.10.0/24</code>	Lists all the hosts scanned (all responding IPs in the subnet from 192.168.10.1 to 192.168.10.254).
<code>nmap -p80,443 192.168.10.10-20</code>	Scans the IP address range looking for open ports 80 and 443.
<code>nmap -p T:80,8080,6588,800 172.16.0.1/22</code>	Scans all hosts between 172.16.0.1 and 172.16.3.254, looking for open TCP ports 80, 8080, 6588 and 800 (the default listening ports for various proxy servers).
<code>nmap -sP 192.168.10.10,20</code>	Ping scans two hosts in a fast scan.
<code>nmap -PN 192.168.10.0/29</code>	Scans all the hosts in the 192.168.10.1 to 192.168.10.6 range. Sometimes, host-based firewalls deny ping requests, and it is difficult to scan such hosts. The -PN scan is useful in such cases; it scans the hosts assuming them to be online.
<code>nmap -A -F 192.168.10.1</code>	Detects target OS and services running on it, in fast-scan mode.

These basic commands are useful for standard scans in any network, and serve a variety of purposes including checking open ports; whether unintended services (like terminal services, VNC, FTP, etc) are running on important hosts; obtaining a list of IP addresses to be scanned, and so on.

However, these simple and straightforward scans may not fulfil all requirements. Sometimes, for example, special scans are required in order to test Intrusion Detection/Prevention systems. There might also be the need to conceal the identity of the scanner from the target host.

Nmap does indeed provide various ways to conceal your IP address (you can also conceal your MAC address by spoofing) though you have to be careful while using these commands. They require an in-depth knowledge of TCP/IP protocols, and may disrupt the systems/network or

CAUTION!

While using Nmap, be warned regarding the following information!

Under the stringent rules of the Indian Cyber Law 2000 and its further amendments till date, even a port scan on a public IP may land you in jail. Do not scan computers that you do not own, or run scans over networks that you do not own, without written permission from the owners.

You may also scan scanme.nmap.org site for testing. This permission only includes scanning via Nmap, and you are not allowed to test exploits and/or denial of service attacks. Don't forget to follow the rules listed in the Nmapman page. Abuse of this service will be reported to the government by the site owners.

Use Nmapvery carefully, and only for discovery/audits of your network. As mentioned above, it is a very powerful tool, and could cause disruption/damage to the target system or network.

Fully understand what you are doing, even with scripted scans, before you do it. Ignorance will not excuse anybody from being prosecuted under the law.

Also note that if you have a website of your own, either hosted at a hosting provider or on a rented physical server, the server and network do NOT belong to you even though you own the website's content. You should ideally obtain permission from such hosting providers/server owners to carry out even 'testing' probes of your own website.

cause damage if not run properly. Let's look at some stealth techniques to conceal the identity of the scanning system.

Idlescan

```
nmap -v -sI 192.168.10.100 192.168.10.105
```

This scan will probe 192.168.10.105 while pretending that the scan packets come from another host; the target's logs will show that the scan originated from 192.168.10.100. This is called a zombie host. In our networking context, zombie hosts are those controlled by other hosts on the network. Not all hosts can be used as zombies, as certain conditions are required to be met before this is possible. (Using packages like *hping* may enable you to find a zombie host on the network.) The switch -v increases the verbosity of the output.

Decoy host

```
nmap -sS -P0 -D 192.168.10.201,192.168.10.202,192.168.10.203 192.168.10.50
```

This command is especially useful while testing IDS/IPS. The -sS option will perform a SYN scan on the target host. While doing so, it will spoof the packet contents